

## Element E

### *Applications of STEM Principles and Practices*

We as a class will be using mathematical calculations, such as those rooted in geometry and trigonometry, to determine different measurements to ensure the best possible design. Especially because there are way more components in the design like the amount of air pressure or the constant flow of water. It's imperative that the team will note down these elements as we conduct our tests, so that if needed, we can adjust our calculations to generate the ideal measurements needed for our hydroponic unit to thrive. This will impact our designing because it'll allow us to make the design scalable, allow us to purchase materials without the worry of waste, and allow us to include features that will better the health of the plants. This will affect our building because it'll allow us to build more efficiently with less error, and permit us to work separately on certain components because of our prior planning. Lastly this will affect our testing because we will use equations to show the effectiveness of the components, and show which ones we will have to alter to make more efficiently. James Roberson told us that our design could use some redesigning; he explained that there were deficiencies in some of the components such as the possibility of water splashing around or out of the compartments due to lack of safeguards. He also recommended us to ask more questions to the stakeholders to find more information on making the hydroponics machine more inclusive to the plants that they want to grow. Mr. Roberson brought up how certain plants may not thrive in the hydroponics machine without some extra accommodations. The team should also know more details on the nutrient source and the water source. How did that impact your process? The team has decided to redo the design in a way that will address all the issues Mr. Roberson brought up. We will be connected with the stakeholders more as well to ensure that all specifications they want are achieved and the hydroponics machine will be beneficial. We chose our final design using an elimination process. This process included considering all the requirements, preferences, and limitations that were given to us by the teacher commissioning this project and through other constraints we were given. The final design hits every point that it should and any drawbacks were considered and acknowledged. They were also communicated to the teachers and experts. The hydroponic units' structure will undergo testing through the process of being filled with water, with the primary objective being to detect any potential leaks that may compromise the functionality of the unit. The presence of leaks poses a significant threat to the success of this project, particularly due to the fact that the unit will be situated in a high-traffic classroom environment where water damage and plant damage could occur as a result. In addition to leak testing, we will also assess the pump force, control pad speed, and make any necessary adjustments to the lighting system. These evaluations will be conducted in a simulated classroom setting, involving tasks such as activating the pump via the control pad, adjusting the light intensity, testing the timer functionality, and assessing the overall mobility of the unit by transporting it across the room. The

information given by our experts and stakeholders about potential obstacles is important because it allows us, as a team, to create a viable solution to present to our consumer.